

11.14 Classwork: Calculus practice (no calculator) — Markscheme

Part A. Differentiate.

1. $y = (3x^2 + 1)^4$

Outer derivative: $4(3x^2 + 1)^3$. Inner derivative: $\frac{d}{dx}(3x^2 + 1) = 6x$. M1

$$\frac{dy}{dx} = 4(3x^2 + 1)^3 \cdot 6x = 24x(3x^2 + 1)^3$$
A1

2. $y = \sqrt{2x^2 + 1} = (2x^2 + 1)^{1/2}$

$$\frac{dy}{dx} = \frac{1}{2}(2x^2 + 1)^{-1/2} \cdot 4x$$
M1

$$= \frac{2x}{\sqrt{2x^2 + 1}}$$
A1

3. $y = \sin(2x^3)$

$$\frac{dy}{dx} = \cos(2x^3) \cdot 6x^2$$
M1

$$= 6x^2 \cos(2x^3)$$
A1

4. $y = e^{x^2 - 4x}$

$$\frac{dy}{dx} = e^{x^2 - 4x} \cdot (2x - 4)$$
M1

$$= (2x - 4)e^{x^2 - 4x}$$
A1

5. $y = \ln(5x^2 + 3)$

$$\frac{dy}{dx} = \frac{1}{5x^2 + 3} \cdot 10x$$
M1

$$= \frac{10x}{5x^2 + 3}$$
A1

6. $y = \cos^3(2x) = [\cos(2x)]^3$

Outer: $3[\cos(2x)]^2$. Then differentiate $\cos(2x)$: $-\sin(2x) \cdot 2 = -2\sin(2x)$. M1 M1

$$\frac{dy}{dx} = 3\cos^2(2x) \cdot (-2\sin(2x)) = -6\cos^2(2x)\sin(2x)$$
A1

Part B. Integrate. For each, identify a substitution u , integrate, then apply the initial condition to find C .

7. $f'(x) = \frac{6x}{\sqrt{x^2 + 4}}, \quad f(0) = 5$

Let $u = x^2 + 4$, so $du = 2x \, dx$ and $6x \, dx = 3 \, du$. M1

$$f(x) = \int \frac{3}{\sqrt{u}} \, du = 6\sqrt{u} + C = 6\sqrt{x^2 + 4} + C \quad \text{A1}$$

Apply $f(0) = 5$: $6\sqrt{4} + C = 5 \Rightarrow 12 + C = 5 \Rightarrow C = -7$ M1 A1

$$f(x) = 6\sqrt{x^2 + 4} - 7 \quad \text{A1}$$

8. $f'(x) = 2x \cos(x^2), \quad f(0) = 1$

Let $u = x^2$, so $du = 2x \, dx$. M1

$$f(x) = \int \cos u \, du = \sin u + C = \sin(x^2) + C \quad \text{A1}$$

Apply $f(0) = 1$: $\sin 0 + C = 1 \Rightarrow C = 1$ M1

$$f(x) = \sin(x^2) + 1 \quad \text{A1}$$

9. $f'(x) = \frac{2x}{x^2 + 1}, \quad f(1) = 0$

Let $u = x^2 + 1$, so $du = 2x \, dx$. M1

$$f(x) = \int \frac{1}{u} \, du = \ln |u| + C = \ln(x^2 + 1) + C \quad \text{A1}$$

Apply $f(1) = 0$: $\ln 2 + C = 0 \Rightarrow C = -\ln 2$ M1

$$f(x) = \ln(x^2 + 1) - \ln 2 = \ln\left(\frac{x^2 + 1}{2}\right) \quad \text{A1}$$

10. $f'(x) = \frac{\ln x}{x}, \quad f(1) = 3$

Let $u = \ln x$, so $du = \frac{1}{x} \, dx$. M1

$$f(x) = \int u \, du = \frac{u^2}{2} + C = \frac{(\ln x)^2}{2} + C \quad \text{A1}$$

Apply $f(1) = 3$: $0 + C = 3 \Rightarrow C = 3$ M1

$$f(x) = \frac{(\ln x)^2}{2} + 3 \quad \text{A1}$$